

Week 5 : Four-axis robotic arm (R4 robotic arm)

Calculation Report

Bill of Material and Assembly Drawings:

ITEM NO.	PART NAME	MATERAILS	QTY
1	arm1	PLA+	1
2	PulleyHtd3m87Tbore8	PLA+	1
3	bearingCover	PLA+	2
4	shaftJ1	PLA+	1
5	Pulleyhtd3m20Tbore8	Aluminum Alloys 6061 Alloy	1
6	pulley2GT80T	Aluminum Alloys 6061 Alloy	1
7	pulley2GT16Tbore6.35	Aluminum Alloys 6061 Alloy	1
8	PulleyHTD3m15Tbore8	Aluminum Alloys 6061 Alloy	1
9	BeltHTD3m312L20T80T	Neoprene	1
10	belt2GT210L	Neoprene	1
11	MagneticSensorV2	MagneticSensorV2	3
12	shaft8x70	PLA+	2
13	Nema17L40PG14	Nema17L40PG14	1
14	Nema23_57x57x56	Nema23_57x57x56	1
15	BeltTensionSpacer	Neoprene	4
16	Washer45x60x1	Steel AISI 304	2
17	NeedleThrustBearing5909_45x60x33	Steel AISI 304	1
18	bearing608 8x22x7	Aluminum Alloys 6061 Alloy	2
19	bearing6806 30x42x7	Steel AISI 304	8
20	Bearing105 5x10x4	Steel AISI 304	16
21	ISO 4762 M4 x 12 - 12N	Steel AISI 304	4
22	ISO 4762 M5 x 25 - 25N	Steel AISI 304	6
23	ISO 4762 M5 x 20 - 20N	Steel AISI 304	2
24	ISO 4762 M4 x 40 - 20N	Steel AISI 304	4
25	ISO 4762 M2 x 12 - 12N	Steel AISI 304	4
26	ISO 4762 M3 x 12 - 12N	Steel AISI 304	4
27	ISO 4762 M5 x 16 - 16N	Steel AISI 304	4
28	ISO 4762 M3 x 16 - 16N	Steel AISI 304	4
29	ISO - 4032 - M5 - W - N	Steel AISI 304	8
30	ISO - 4032 - M2 - W - N	Steel AISI 304	8
31	ISO - 4032 - M4 - W - N	Steel AISI 304	4
32	BeltHTD3m363L15T87T	Neoprene	1

ITEM NO.	PART NAME	MATERAILS	QTY
33	Base	PLA+	1
34	pulleyHTD3m67Tbase	PLA+	1
35	shaftJ1endPlate	PLA+	1
36	MagneticSensor	MagneticSensor	1
37	arm2	PLA+	1
38	spacerM3x20	PLA+	8
39	Nema 17 model v2	Nema 17 model v2	2
40	bearing608 8x22x7	Steel AISI 304	2
41	pulley2GT82Tbore8	Aluminum Alloys 6061 Alloy	1
42	pulley2GT17Tbore5	Aluminum Alloys 6061 Alloy	1
43	belt2GT310L	Neoprene	1
44	PulleyHtd3m67T	PLA+	1
45	ISO 4762 M2 x 12 - 12N	Steel AISI 304	4
46	ISO 4762 M4 x 40 - 20N	Steel AISI 304	8
47	ISO 4762 M3 x 30 - 18N	Steel AISI 304	8
48	ISO 4762 M5 x 25 - 25N	Steel AISI 304	4
49	ISO 4762 M4 x 12 - 12N	Steel AISI 304	4
50	BeltHTD3m420L14T67T	Neoprene	1
51	PulleyHTD3m14Tbore8	Aluminum Alloys 6061 Alloy	1
52	ISO - 4035 - M5 - N	Steel AISI 304	4
53	arm3	PLA+	1
54	Pulley2GT70T	PLA+	1
55	pulley2GT20Tbore5	Aluminum Alloys 6061 Alloy	1
56	belt2GT520L	Neoprene	1
57	gripper holder	PLA+	1
58	Gripper	Gripper	1
59	Payload	PLA+	1

Isometric view:

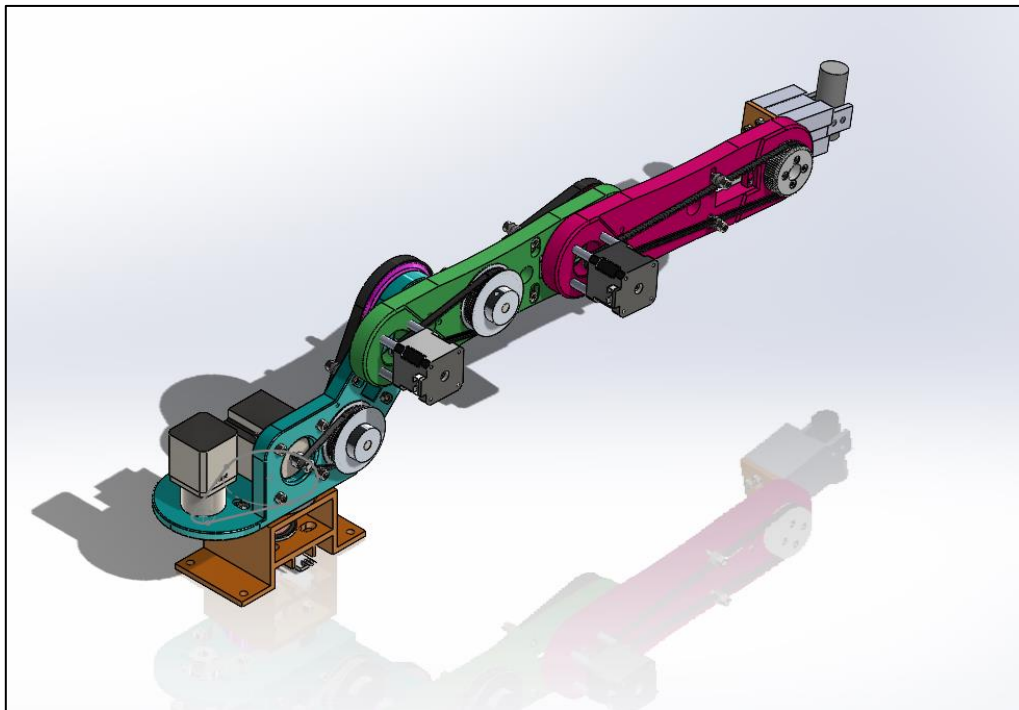


Fig. 1 R4 Robotic Arm

Explode view:

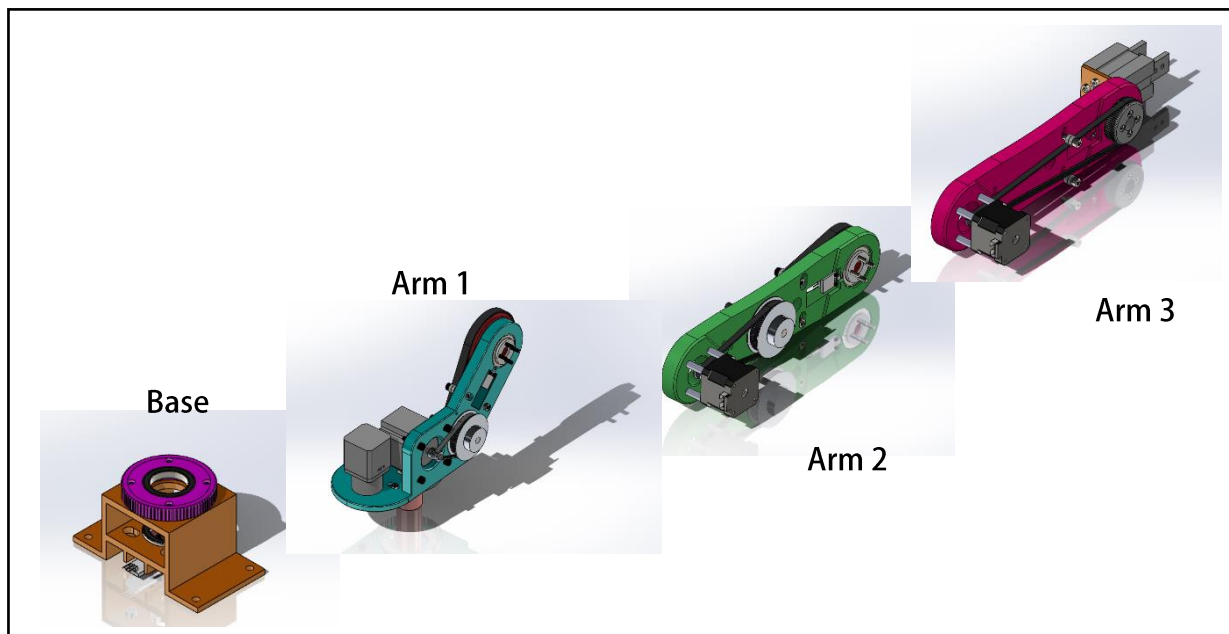


Fig. 2 Explode view 1

Arm3 Assembly:

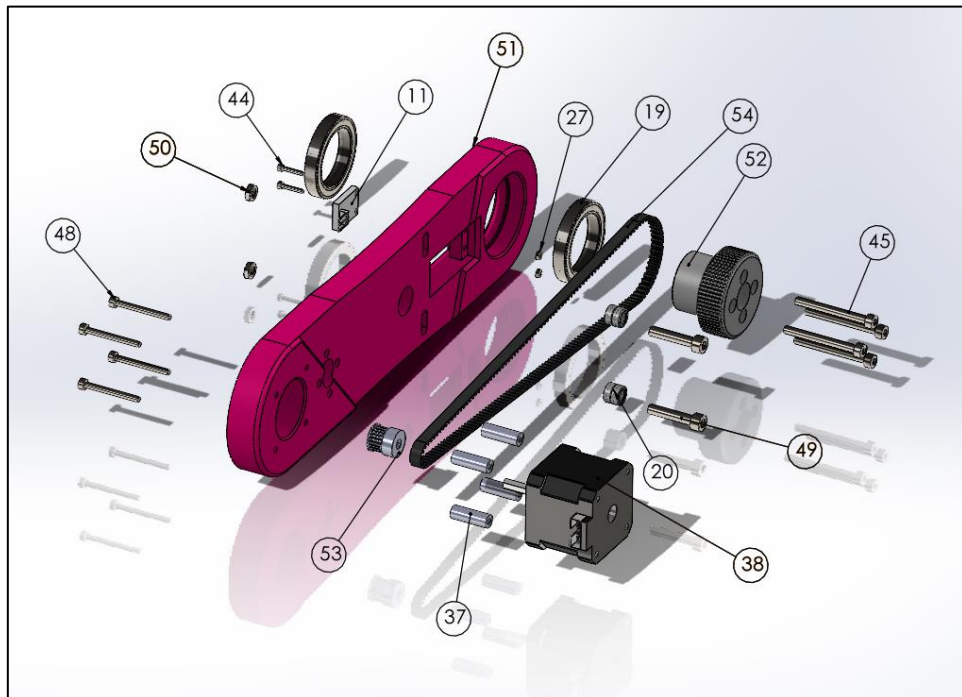


Fig. 3 Arm3 assembly

Arm2 Assembly:

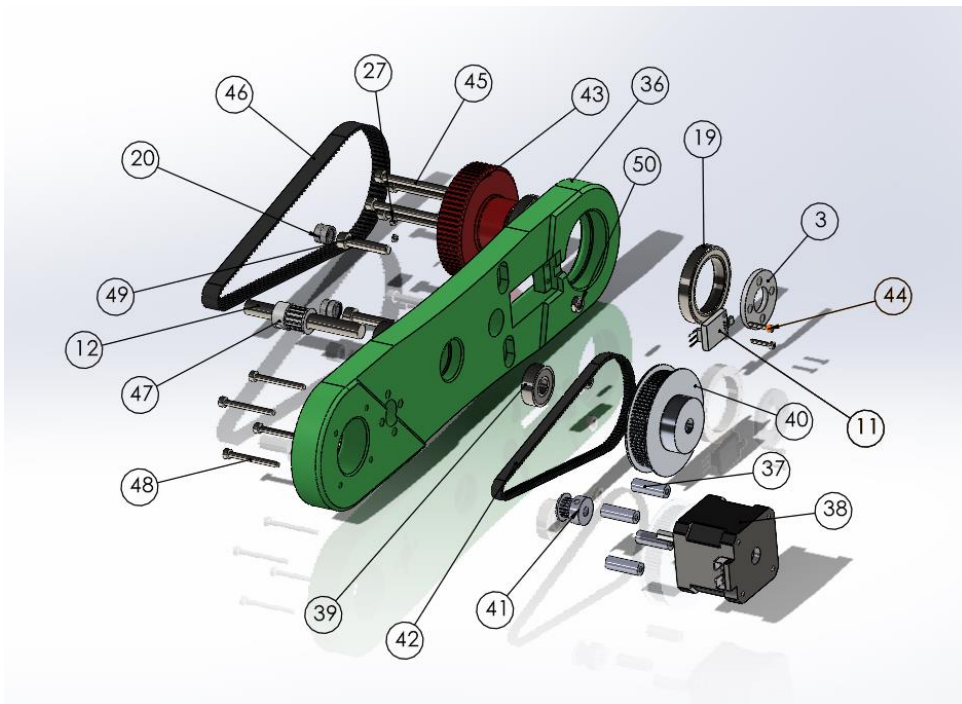


Fig. 4 Arm2 assembly

Arm1 Assembly:

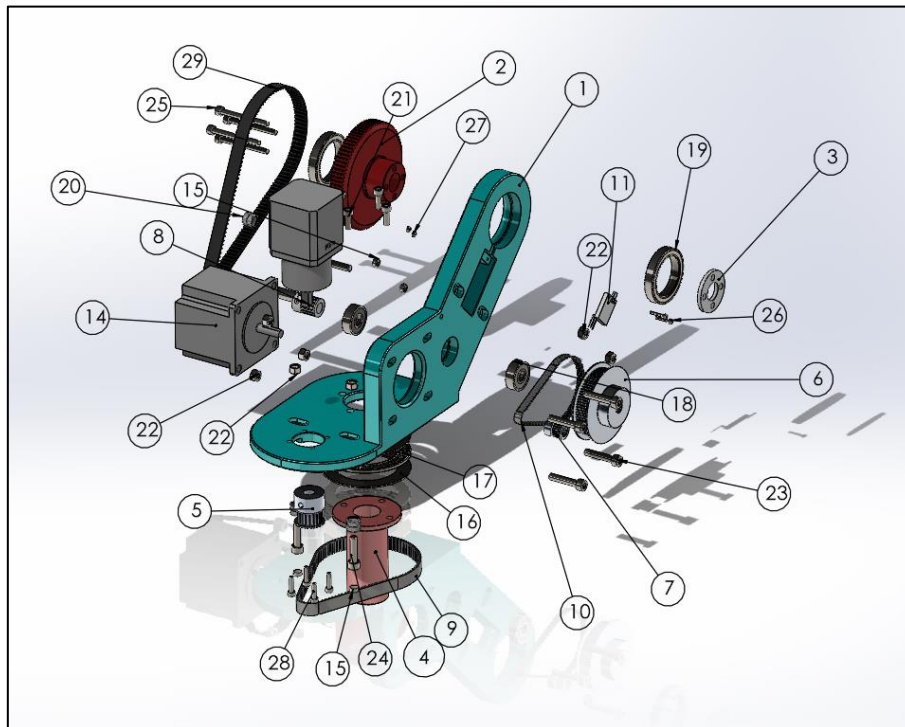


Fig. 5 Arm1 assembly

Base Assembly:

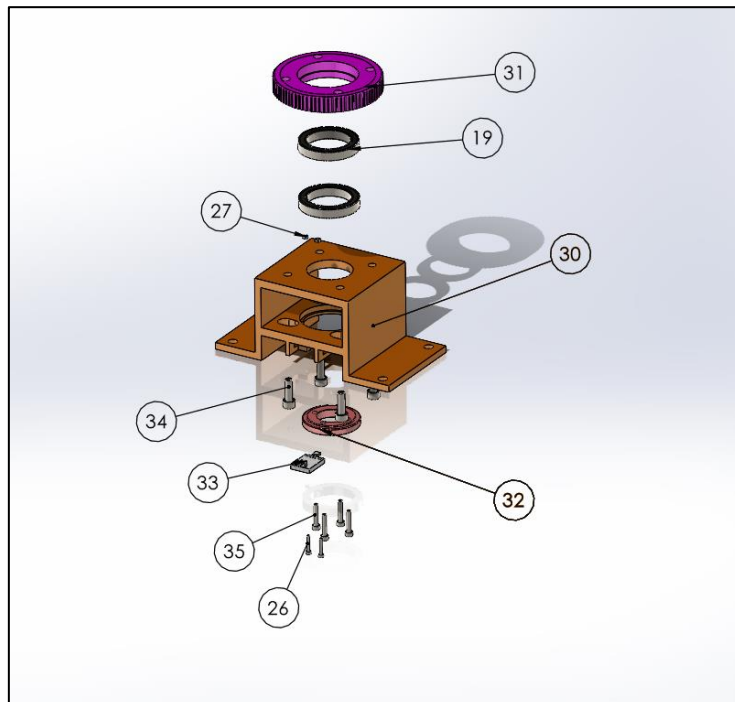


Fig. 6 Base assembly

Joint Configuration :

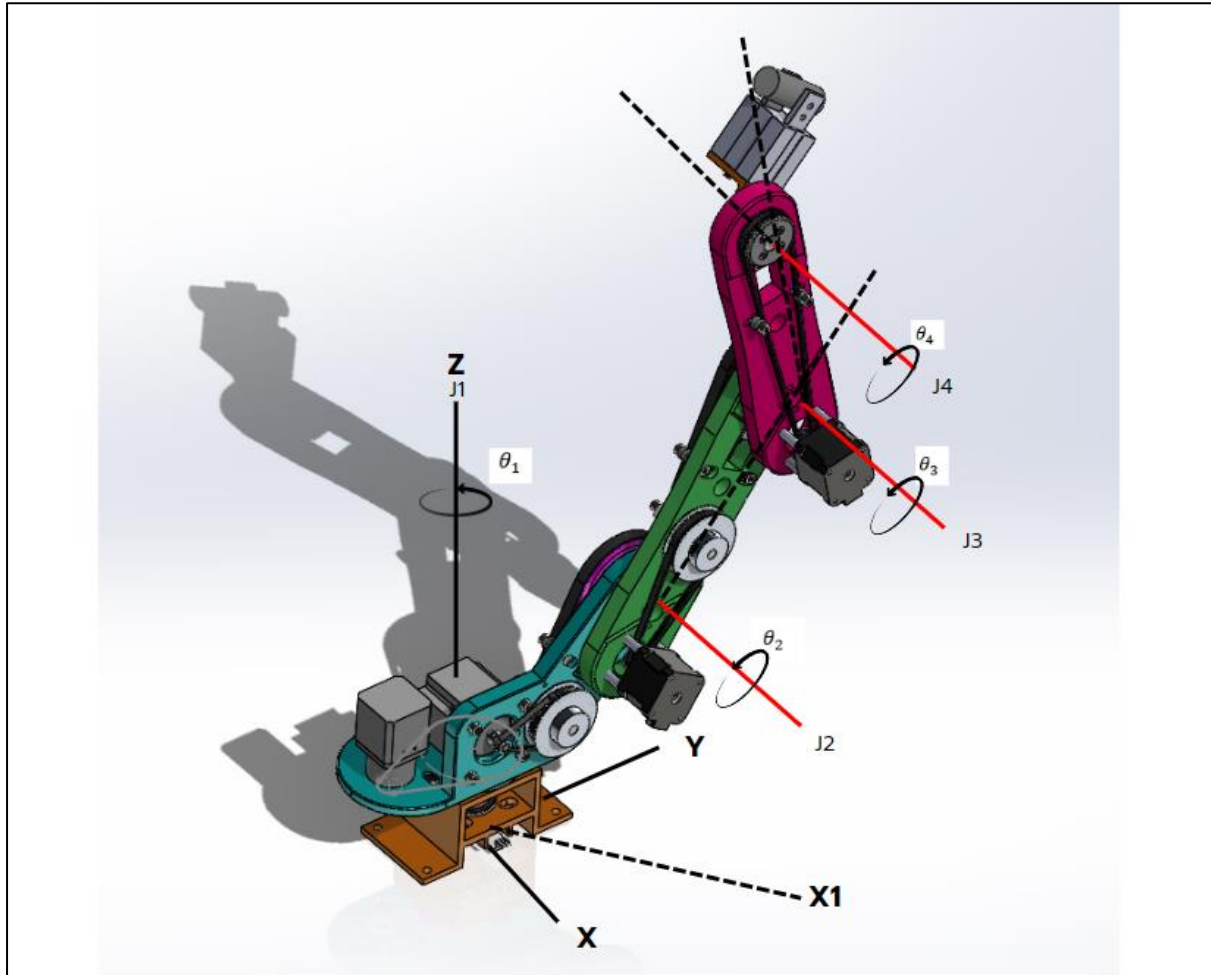
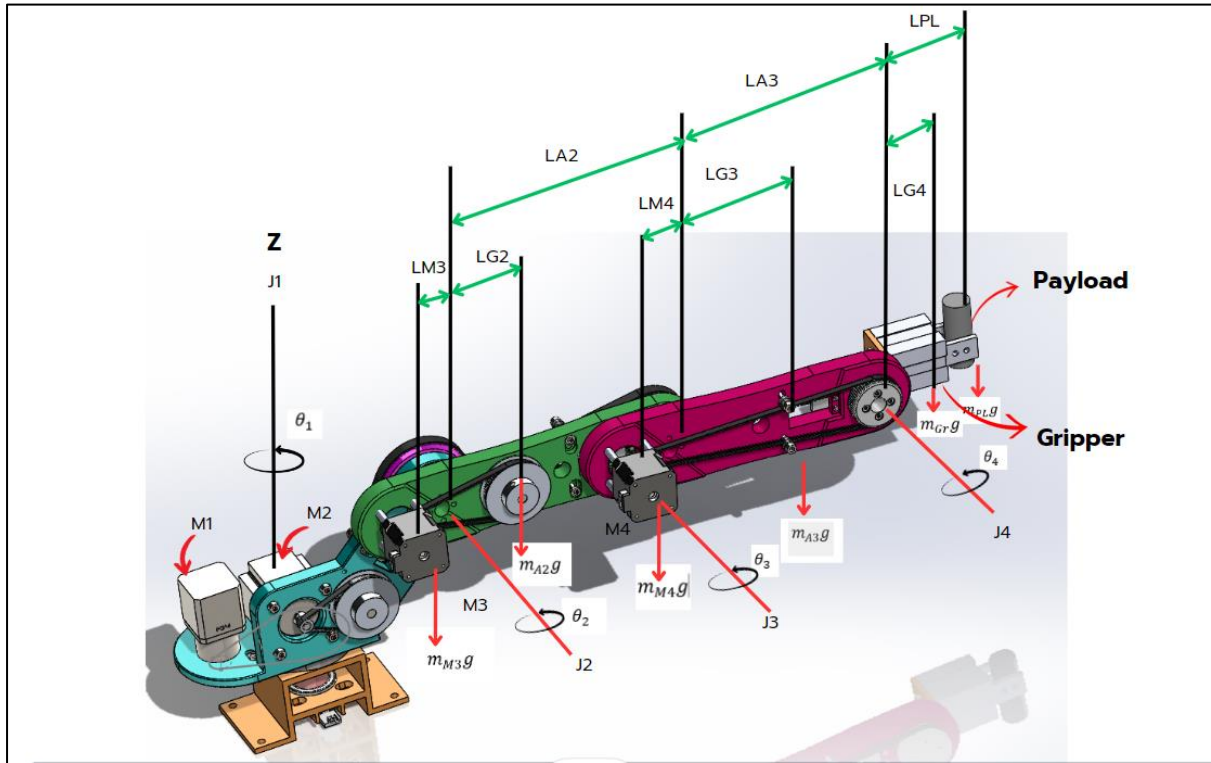


Fig. 7 Configuration of R4 robotic arm joints

Load Torque:



From Fig. 8, the load torque can be expressed as

$$T_{L,J4} = g[m_{Gr} + L_{G4} + m_{PL}L_{PL}]$$

$$T_{L,J3} = g[m_{M4} + L_{M4} + m_{A3}L_{A3} + m_{Gr}(L_{G4} + L_{A3}) + m_{PL}(L_{PL} + L_{A3})]$$

$$T_{L,J2} = g[m_{M3}(-L_{M3}) + m_{A2}(L_{A2}) + m_{M4}(-L_{M4} + L_{A2}) + m_{PL}(L_{A3} + L_{A2}) + m_{Gr}(L_{G4} + L_{A3} + L_{A2}) + m_{PL}(L_{PL} + L_{A3} + L_{A2})]$$

Where $T_{L,J2}$ is the load torque of joint J_2 , $T_{L,J3}$ is the load torque of joint J_3 , $T_{L,J4}$ is the load torque of joint J_4 , m_{M3} motor M3 mass, m_{M4} is the motor M4 mass, m_{A2} is the arm 2 mass, m_{A3} is the arm 3 mass, m_{Gr} is the gripper mass, and m_{PL} is the payload mass.

Given:

$$m_{PL} = 500g$$

$$m_{M1} = 800g \text{ (Nema17 stepper motor with planetary gearbox 5.18:1)}$$

$$m_{M2} = 640g \text{ (Nema23 stepper motor)}$$

$$m_{M3} = 520g \text{ (Nema17 stepper motor)}$$

$$L_{A2} = 197.51 \text{ mm}, L_{A3} = 180 \text{ mm}, L_{PL} = 0.085 \text{ m}, L_{M3} = 40 \text{ mm}, L_{M4} = 40 \text{ mm}$$

Use the following link to help you determine the mass of each arm and its center of mass.

The final values we obtained are:

$$m_{A1} = 595.73 \text{ g}, m_{A2} = 501.88 \text{ g}, m_{A3} = 291.41 \text{ g}, m_{Gr} = 322.90 \text{ g}, L_{G2} = 1.63 \text{ mm}, L_{G3} = 2.46 \text{ mm}, \text{ and } L_{G4} = 22.03 \text{ mm}.$$

At joint J1, a thrust bearing is used with two washers as shown in Fig. 9. The load torque at joint J1 can be written as

where m is the total mass mouthed on the thrust bearing, $T_{L,J1}$ is the load torque of joint J_1 , μ is the friction coefficient of the thrust bearing (approximately 0.1), R_o is the outer diameter of the thrust bearing, and R_i is the inner diameter of the thrust bearing. The total mass can be calculated from

$$= 595.73 + 501.88 + 291.41 + 800 + 640 + 520 + 520 + 322.90 + 500 \\ = 4691.92 \text{ g}$$

A thrust bearing 5909 45x60x3 with a 45 mm inner diameter and a 60 mm outer diameter is used.

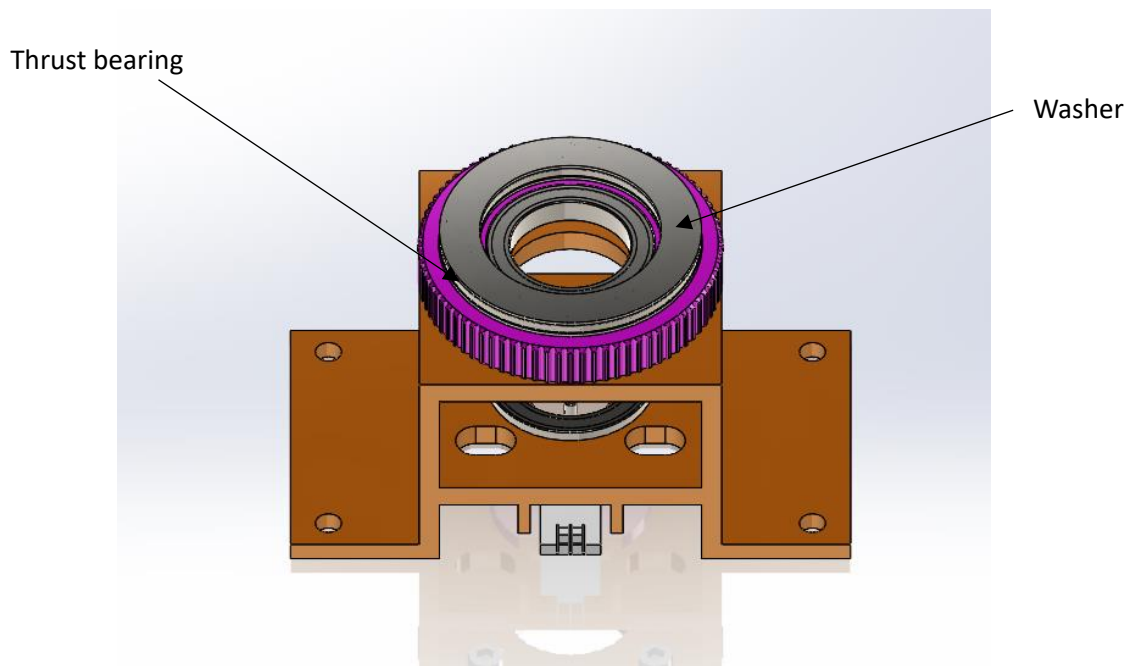


Fig. 9 Thrust bearing at joint J_1

$$T_{L,J4} = 9.81[322.90(22.03) + 500(100)] \times 10^{-9} = 0.56 \text{ N.m}$$

$$T_{L,J3} = 9.81[520(-40) + 291.41(180) + 322.90(22.03 + 180) + 500(0.085 + 180)] \times 10^{-9} = 1.834 \text{ N.m}$$

$$T_{L,J2} = 9.81[520(-40) + 501.88(197.51) + 520(-40 + 197.51) + 291.41(180 + 197.51) + 322.90(22.03 + 180 + 197.51) + 500(100 + 180 + 197.51)] \times 10^{-9} = 6.259 \text{ N.m}$$

$$T_{L,J1} = \frac{2}{3}(0.1)(4691.92 * 10^{-3})(9.81) \frac{0.06^3 - 0.045^3}{0.06^2 - 0.045^2} = 0.243 \text{ N.m}$$

Acceleration torque

When the motor accelerates from zero speed to the desired speed over time t , the acceleration and deceleration for all joints can be expressed as

$$\alpha = \frac{\omega}{t} = \frac{2\pi n}{60t}$$

Use $t=1$ s and $n = 20$ rpm for all joints

$$\alpha = \frac{2\pi(20)}{60(1)} = 2.09 \text{ rad/s}^2$$

The acceleration torque can be express as

$$T_{0,J4} = I_{J4}\alpha$$

$$T_{0,J3} = I_{J3}\alpha$$

$$T_{0,J2} = I_{J2}\alpha$$

$$T_{0,J1} = I_{J1}\alpha$$

Use the link below to determine the mass moment of inertia about any joints of the robotic arm using SOLIDWORKS. Finally, we found that,

$$I_{J4} = 1058386.12 * 10^{-9} \text{ kg/m}^3$$

$$I_{J3} = 4263832.09 * 10^{-9} \text{ kg/m}^3$$

$$I_{J2} = 6665653.88 * 10^{-9} \text{ kg/m}^3$$

$$I_{J1} = 386519421.17 * 10^{-9} \text{ kg/m}^3$$

$$T_{0,J4} = (1058386.12 * 10^{-9})(2.09) = 0.002 \text{ N.m}$$

$$T_{0,J3} = (4263832.09 * 10^{-9})(2.09) = 0.009 \text{ N.m}$$

$$T_{0,J2} = (6665653.88 * 10^{-9})(2.09) = 0.014 \text{ N.m}$$

$$T_{0,J1} = (386519421.17 * 10^{-9})(2.09) = 0.808 \text{ N.m}$$

Required torque

The require torque can be determined from

$$T = SF(T_L + T_a)$$

Use a safety factor of $SF = 1.7$

$$T_{J4} = 1.7(0.56 + 0.002) = 0.956 \text{ N.m}$$

$$T_{J3} = 1.7(1.834 + 0.009) = 3.133 \text{ N.m}$$

$$T_{J2} = 1.7(6.259 + 0.014) = 10.664 \text{ N.m}$$

$$T_{J1} = 1.7(0.243 + 0.808) = 1.788 \text{ N.m}$$

Motor selection and required gear ratio

Joint J_4 and J_3 :

HANPOSE - Stepper motor (Nema17), Holding torque of 0.70 N.m, 1.7A (17HS6401S / 42HS60) Black End,

motor mass: $m_{M3} = m_{M3} = 0.52 \text{ kg}$.

<https://www.zonemaker.com/product/190/hanpose-stepper-motor-nema17-%E0%B9%81%E0%B8%A3%E0%B8%87%E0%B8%9A%E0%B8%B4%E0%B8%94-70-n-cm-1-7a-17hs6401s-black-end>

$$T_{M3} = T_{M4} = 0.4(0.7) = 0.280 \text{ N.m}$$

$$i_{required} = \frac{T}{T_M}$$

$$i_{4,required} = \frac{T_{J4}}{T_{M4}} = \frac{0.956}{0.280} = 3.41:1$$

$$i_{3,required} = \frac{T_{J3}}{T_{M3}} = \frac{3.133}{0.280} = 11.19:1$$

Joint J_2 :

HANPOSE - Nema23 Stepper Motor, Holding torque of 1.01N.m, 2.8A, Shaft 6.35mm
(23HS5128-6.35) motor mass: $m_{M2} = 0.64 \text{ kg}$

<https://www.zonemaker.com/product/3045/hanpose-stepper-motor-nema23-%E0%B9%81%E0%B8%A3%E0%B8%87%E0%B8%9A%E0%B8%B4%E0%B8%94-1-01n-m-2-8a-shaft-6-35mm-23hs5128-6-35>

$$T_{M2} = 0.4(1.01) = 0.404 \text{ N.m}$$

$$i_{2,required} = \frac{T_{J2}}{T_{M2}} = \frac{10.664}{0.404} = 26.40:1$$

Joint J_1 :

HANPOSE-Planetary Gearbox Ratio 5.18:1 Nema17 Stepper Motor (17HS2408S-PG5.18):
holding torque of 0.4 N.m. motor and planetary gear box mass: $m_{M2} = 0.8 \text{ kg}$

<https://www.zonemaker.com/product/744/hanpose-planetary-gearbox-ratio-5-181-nema17-stepper-motor-17hs4401s-pg5-18>

$$T_M = 0.4T_{M,max}$$

$$T_{M2} = 0.4(0.4) = 0.160 \text{ N.m}$$

$$i_{1,required} = \frac{T_{J1}}{T_{M1}} = \frac{1.788}{0.160} = 11.18:1$$

Power transmission selection

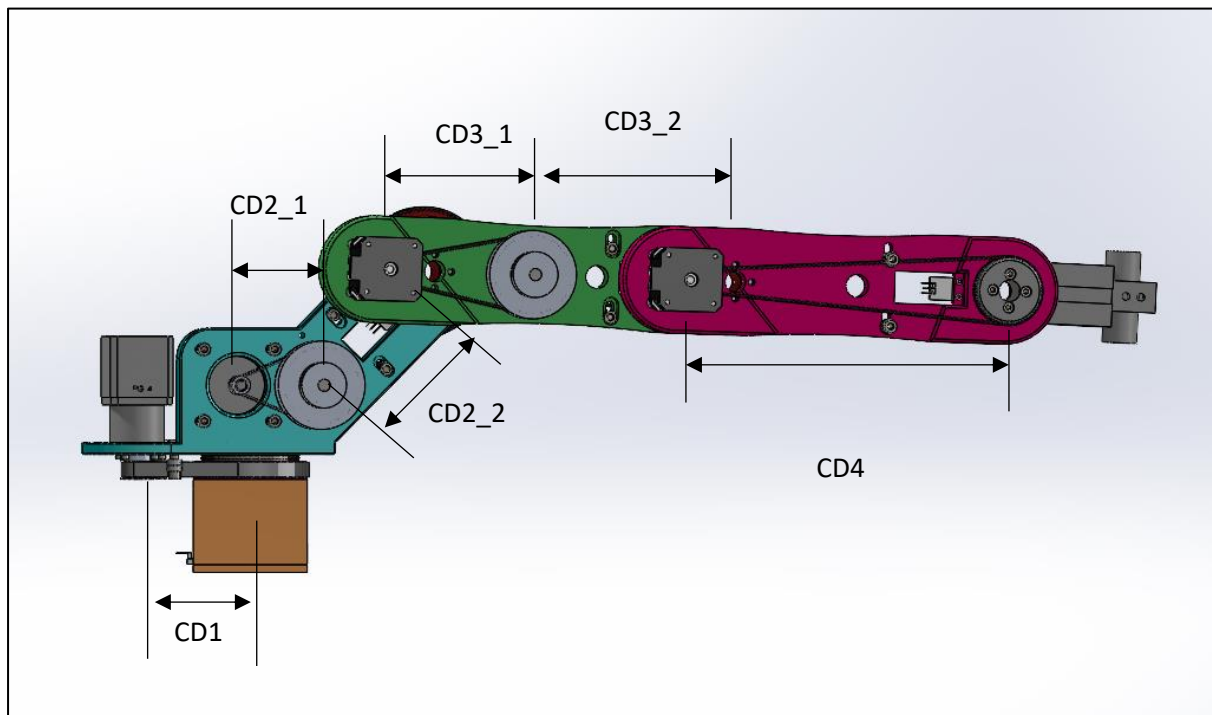


Fig. 10

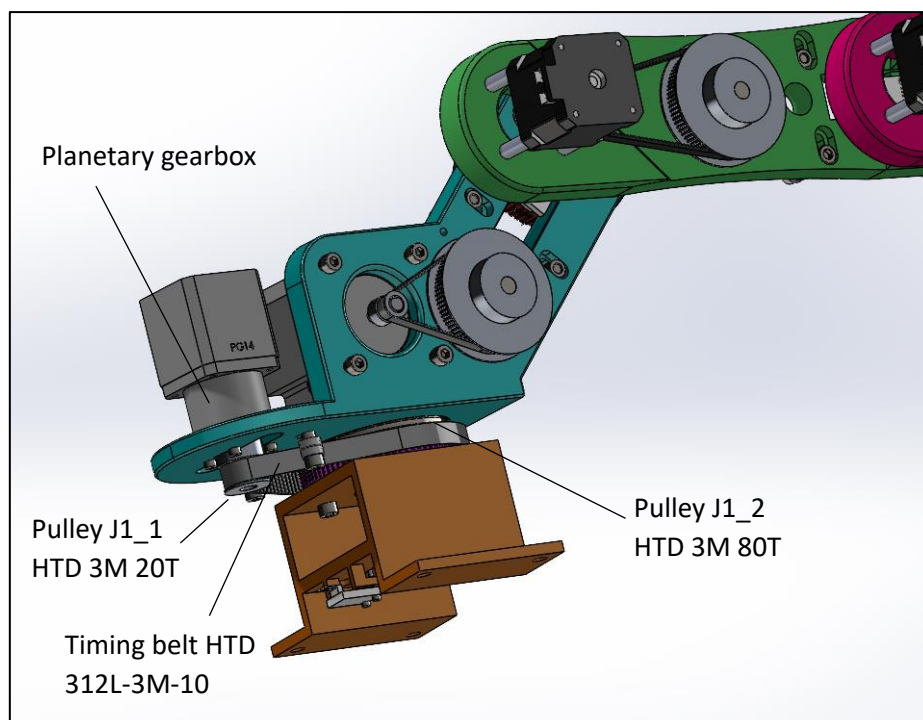


Fig. 11

Stage 1: Planetary gearbox, $i_{j1/1} = 5.18:1$

Stage 2:

HTD-3M timing belt, 10 mm wide, with a closed loop length of 312 mm.

<https://www.zonemaker.com/product/618/htd-3m-width-10mm-closed-loop-belt-length-312mm>

Pulley 1_1, HTM-3M, 20 teeth, 8 mm bore.

<https://www.zonemaker.com/product/622/htd-3m-10mm-20-teeth-timing-pulley-bore-8mm>

Pulley 1_2, HTM-3M, 80 teeth, 3D printed.

$$i_{1_1} = \frac{N_{1_2}}{N_{1_1}} = \frac{80}{20} = 4:1$$

$$i_1 = i_{1_1} \cdot i_{1_2} = (5.18)(4) = 20.72:1 > i_{1,required} (11.18:1)$$

A gear ratio of $i_1 = 20.72:1$ can be used to operate the robotic arm's joint J_1 .

A pitch diameter of the pulley can be determined by

$$D = pN/\pi \quad (HTD\ 3M, p = 3\ mm)$$

$$D = \frac{3(20)}{\pi} = 19.10\ mm$$

$$D = \frac{3(80)}{\pi} = 76.39\ mm$$

The center distance of the pulleys can be determined by

$$b = 2PL - \pi(D_1 + D_2)$$

$$b = 2(312) - \pi(19.10 + 76.39) = 324\ mm$$

$$CD = \frac{b\sqrt{b^2 - 8(D_2 - D_1)^2}}{8}$$

$$CD_1 = \frac{324\sqrt{324^2 - 8(76.39 - 19.10)^2}}{8} = 75.6\ mm$$

Joint J_2

Stage 1:

2GT Timing belt, 6 mm wide, with a closed loop length of 210 mm.

<https://shopee.co.th/%F0%9F%94%A5-%E0%B8%9E%E0%B8%A3%E0%B9%89%E0%B8%AD%E0%B8%A1%E0%B8%AA%E0%B9%88%E0%B8%87-%F0%9F%94%A5%E0%B8%AA%E0%B8%B2%E0%B8%A2%E0%B8%9E%E0%B8%B2%E0%B8%99-2gt-6-%E0%B9%81%E0%B8%9A%E0%B8%9A%E0%B8%9B%E0%B8%B4%E0%B8%94-%E0%B9%80%E0%B8%AA%E0%B9%89%E0%B8%99%E0%B8%A3%E0%B8%AD%E0%B8%9A%E0%B8%A7%E0%B8%87-200-298mm.Timing-belt-width-6mm-%E0%B9%81%E0%B8%9A%E0%B8%9A-close-loop-%E0%B9%80%E0%B8%AA%E0%B9%89%E0%B8%99%E0%B8%A3%E0%B8%AD%E0%B8%9A%E0%B8%A7%E0%B8%87-200-298mm.-i.252491788.9463831265?xptdk=b3caed6c-7e29-43c6-8929-5e622752be9e>

Pulley 2_1, 2GT, 16 teeth, 6.35 mm bore.

<https://www.zonemaker.com/product/721/2gt-width-6mm-16-teeth-timing-pulley-bore-6-35mm>

Pulley 2_2, 2GT, 80 teeth, 8 mm bore.

<https://shopee.co.th/%F0%9F%94%A5-%E0%B8%9E%E0%B8%A3%E0%B9%89%E0%B8%AD%E0%B8%A1%E0%B8%AA%E0%B9%88%E0%B8%87-%F0%9F%94%A5-Pulley-2GT%E0%B8%9E%E0%B8%B9%E0%B9%80%E0%B8%A5%E0%B9%88-80%E0%B8%9F%E0%B8%B1%E0%B8%99-2GT-6-80-teeth-Timing-Pulley-BF-%E0%B8%AB%E0%B8%99%E0%B9%89%E0%B8%B2%E0%B8%81%E0%B8%A7%E0%B9%89%E0%B8%B2%E0%B8%87%E0%B8%AA%E0%B8%B2%E0%B8%A2%E0%B8%9E%E0%B8%B2%E0%B8%99-6-10-mm.-i.252491788.22318651118?xptdk=15cf7b14-b5c4-4dce-9e98-88e1b6cfbb9c>

$$i_{2-1} = \frac{N_{22}}{N_{21}} = \frac{80}{16} = 5:1$$

A pitch diameter of the pulley can be determined by

$$D_{2_1} = 2(16)/\pi = 10.19 \text{ mm} \quad (2GT, p = 2 \text{ mm})$$

$$D_{2_2} = 2(80)/\pi = 50.93 \text{ mm}$$

The center distance of the pulleys can be determined by

$$b = 2(210) - \pi(10.19 + 50.93) = 228 \text{ mm}$$

$$CD_{2_1} = \frac{228\sqrt{228^2 - 8(50.93 - 10.19)^2}}{8} = 75.6 \text{ mm}$$

Stage 2:

HTD-3M timing belt, 10 mm wide, with a closed loop length of 363 mm.

<https://www.zonemaker.com/product/1920/htd-3m-width-10mm-closed-loop-belt-length-363mm>

Pulley 2_3, HTM-3M, 15 teeth, 8 mm bore.

<https://www.zonemaker.com/product/621/htd-3m-10mm-15-teeth-timing-pulley-bore-8mm>

Pulley 3_4, HTM-3M, 87 teeth, 3D printed.

$$i_{2_2} = \frac{N_{2_4}}{N_{2_3}} = \frac{87}{15} = 5.8:1$$

A pitch diameter of the pulley can be determined by

$$D_{2_3} = \frac{15(3)}{\pi} = 14.32 \text{ mm} \quad (HTD 3m, p = 3 \text{ mm})$$

$$D_{2_4} = \frac{87(3)}{\pi} = 83.08 \text{ mm}$$

The center distance of the pulleys can be determined by

$$b = 2(363) - \pi(83.08 + 14.32) = 420.16$$

$$CD = \frac{420.16 + \sqrt{420.16^2 - 8(83.08 - 14.32)^2}}{8} = 99.07 \text{ mm}$$

$$i_2 = i_{2_2} = (5)(5.8) = 29:1 > i_{2,required} (26.40:1)$$

A gear ratio of $i_2 = 29:1$ can be used to operate the robotic arm's joint J_2

Joint J_3

Stage 1:

2GT Timing belt, 6 mm wide, with a closed loop length of 300 mm.

<https://www.zonemaker.com/product/2166/belt-closed-loop-rubber-300-2gt-6-closed-loop-length-300mm-gt2-timing-belt-width-6mm-2>

Pulley 3_1, 2GT, 17 teeth, 5 mm bore.

<https://shopee.co.th/%F0%9F%94%A5-%E0%B8%9E%E0%B8%A3%E0%B9%89%E0%B8%AD%E0%B8%A1%E0%B8%AA%E0%B9%88%E0%B8%87-%F0%9F%94%A5Pulley-%E0%B8%9E%E0%B8%B9%E0%B9%80%E0%B8%A5%E0%B9%88-%E0%B8%9E%E0%B8%A5%E0%B8%B9%E0%B9%80%E0%B8%A5%E0%B9%88-2gt-20-%E0%B8%9F%E0%B8%B1%E0%B8%99-16-%E0%B8%9F%E0%B8%B1%E0%B8%99-%E0%B8%AB%E0%B8%99%E0%B9%89%E0%B8%B2%E0%B8%81%E0%B8%A7%E0%B9%89%E0%B8%B2%E0%B8%87%E0%B8%AA%E0%B8%B2%E0%B8%A2%E0%B8%9E%E0%B8%B2%E0%B8%99-6mm-.Pulley2GT20teeth-i.252491788.4833675066?xptdk=b41d5e09-b5d4-41ef-b3d9-6e628952ea57>

Pulley 3_2, 2GT, 82 teeth, 8 mm bore.

<https://shopee.co.th/%F0%9F%94%A5-%E0%B8%9E%E0%B8%A3%E0%B9%89%E0%B8%AD%E0%B8%A1%E0%B8%AA%E0%B9%88%E0%B8%87-%F0%9F%94%A5-Pulley-2GT%E0%B8%9E%E0%B8%B9%E0%B9%80%E0%B8%A5%E0%B9%88-80%E0%B8%9F%E0%B8%B1%E0%B8%99-2GT-6-80-teeth-Timing-Pulley-BF-%E0%B8%AB%E0%B8%99%E0%B9%89%E0%B8%B2%E0%B8%81%E0%B8%A7%E0%B9%89%E0%B8%B2%E0%B8%87%E0%B8%AA%E0%B8%B2%E0%B8%A2%E0%B8%9E%E0%B8%B2%E0%B8%99-6-10-mm.-i.252491788.22318651118?xptdk=15cf7b14-b5c4-4dce-9e98-88e1b6cfbb9c>

$$i_{3,1} = \frac{N_{3,2}}{N_{3,1}} = \frac{82}{17} = 4.82:1$$

A pitch diameter of the pulley can be determined by

$$D_{3,1} = \frac{17(2)}{\pi} = 10.82 \text{ mm} \quad (2GT, p = 2 \text{ mm})$$

$$D_{3,2} = \frac{82(2)}{\pi} = 52.2 \text{ mm}$$

The center distance of the pulleys can be determined by

$$b = 2(310) - \pi(52.2 + 10.82) = 422.02$$

$$CD = \frac{422.02 + \sqrt{422.02^2 - 8(52.2 - 10.82)^2}}{8} = 103.44mm$$

Stage 2:

HTD-3M timing belt, 10 mm wide, with a closed loop length of 417 mm.

<https://www.zonemaker.com/product/2444/htd-3m-width-10mm-closed-loop-belt-length-417mm>

Pulley 3_3, HTM-3M, 14 teeth, 8 mm bore.

<https://www.zonemaker.com/product/622/htd-3m-10mm-20-teeth-timing-pulley-bore-8mm>

Pulley 3_4, HTM-3M, 67 teeth, 3D printed.

$$i_{3_2} = \frac{N_{3_2}}{N_{3_1}} = \frac{67}{14} = 4.79: 1$$

A pitch diameter of the pulley can be determined by

$$D_{3_3} = \frac{14(3)}{\pi} = 13.37mm \quad (HTD\ 3m, p = 3\ mm)$$

$$D_{3_4} = \frac{82(3)}{\pi} = 78.3mm$$

The center distance of the pulleys can be determined by

$$b = 2(420) - \pi(78.3 + 13.37) = 552.01$$

$$CD = \frac{552.01 + \sqrt{552.01^2 - 8(78.3 - 13.37)^2}}{8} = 134.07mm$$

$$i_3 = i_{3_2} = (4.82)(4.79) = 23.09: 1 > i_{3,required} (11.19: 1)$$

A gear ratio of $i_3 = 23.09: 1$ can be used to operate the robotic arm's joint J_3

Joint J4:

2GT Timing belt, 6 mm wide, with a closed loop length of 500 mm.

<https://www.zonemaker.com/product/2168/belt-closed-loop-rubber-500-2gt-6-closed-loop-length-500mm-gt2-timing-belt-width-6mm-2>

Pulley 4_1, 2GT, 20 teeth, 5 mm bore.

<https://www.zonemaker.com/product/65/2gt-width-6mm-20-teeth-timing-pulley-bore-5mm>

Pulley 4_2, 2GT, 70 teeth, 3D printed.

$$i_{4_1} = \frac{N_{4_2}}{N_{4_1}} = \frac{70}{20} = 3.5:1 > i_{4,required} \text{ (3.41:1)}$$

A gear ratio of $i_4 = 3.5:1$ can be used to operate the robotic arm's joint J_4